

Location Representations for Accelerating the Training of Next POI Recommender System

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ABSTRACT

With the widespread use of mobile devices, many next Point-Of-Interest (POI) recommender systems train users' data on their devices to provide personalized recommendation services, which requires the algorithms to have an efficient convergence rate. Besides, POI location, an important feature in improving the POI recommendation performance, has been represented in various ways, such as encoding locations with distinctive IDs (e.g., different areas with different IDs). However, the positional relationship of a POI of interest and other POIs is ignored, which is crucial to predicting the next POI. In this study, we propose describing a POI's location representation as the positional relationship with other POIs. Based on the idea that users prefer to visit closer regions, our approach transfers the distance interval of POIs to the similarity of the location representations, thus the possibility of a user visiting a POI can be revealed from the representation similarities. The proposed location representations that contain relative location information of all POIs can be applied to initialize POI recommendation models to accelerate model convergence. Experiments on our approach show that the proposed method on a sequential recommendation model improves Hit@10 by 4%, and the convergence rate (average regret before epoch@150) increases 4% with a 10% Hit@10 improvement.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile devices; Social networking sites*; • **Information systems** → *Collaborative filtering; Collaborative filtering*.

KEYWORDS

Next POI recommendation, convergence rate, POI location

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1 INTRODUCTION

The next POI recommender system has been used on various location-aware applications. In particular, the location is an important attribute of POI, which directly relates to the current location of a user and thus improves the recommendation performance. Existing studies [6, 8] utilize the POI locations in different ways. For example, Cui et al. [1] groups POIs in close areas by Geohash IDs (an algorithm that divides the map into grids and encodes grids with different IDs) and concatenates Geohash embedding (a series of trainable parameters in the model) to POI embedding for model training. However, these methods consider position relationships within a small amount of POIs while ignoring the relationships to all other POIs. For instance, given two grids, the above methods can only represent the relationships of POIs within each grid while ignoring the relationships of POIs between the grids. Besides, many recommender systems deployed on user devices require user data for training to provide personalized recommendation services. Thus, the model's convergence rate becomes an important factor in measuring system performance. [2, 7] try to accelerate the convergence rate by modifying the optimization functions, however the convexity assumptions in the study may not hold in practice.

This raises the question if it is possible to accelerate the model convergence rate and improve recommendation performance simultaneously by finding a good representation of POI locations? To address the problem, we design an algorithm to represent the positional relationship of each POI to other POIs and then equip the representations to the next POI recommendation models. To the best of our knowledge, it is the first attempt to employ positional relationships of POIs to improve model performance and accelerate model convergence simultaneously. Our work makes two major contributions: (1) Our approach introduces a good representation of POI locations that contains the positional relationships with all other POIs. The location representation improves the recommendation performance in various models. (2) The design of the location representations can help the model reach an earlier convergence.

2 METHODS

Our approach aims to learn POI representations containing global position relationships. We first divide the map into P grids, POIs that are located in the same grid are supposed to have the same location representations (this step aims to address data sparsity issue). We model the representation of each grid p to every target grid t ($t \in P$) using:

$$g_{p,t} = \frac{-1}{1 + e^{-d_{man}(p,t)}}, \quad (1)$$

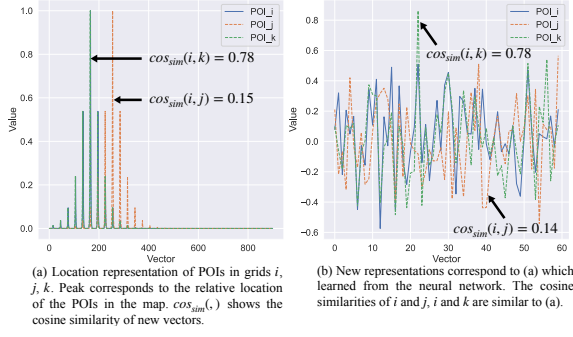


Figure 1: An example of location representations and the corresponding new features calculated by the network.

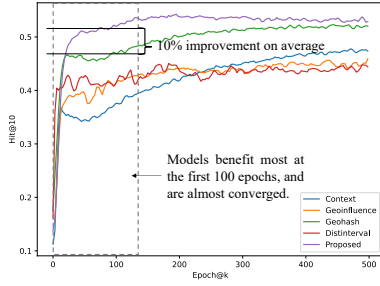


Figure 2: Visualize of the convergence rate of all sequential models.

where $d_{man}(\cdot)$ denotes the distance between two grids, for simplicity, we apply Manhattan distance. The representation of p is organized as $S_p = [g_{p,1}, g_{p,2}, \dots, g_{p,|P|}]$. $g_{p,t}$ ranges from 0 to 1, the closer t and p are, the greater the value of $g_{p,t}$. Because S_p includes not only the position of POI in the map (the location of peak in Figure 1 (a)) but also the relative distance to other POIs (cos similarity), as shown in Figure 1 (a), the recommender system can thus predict net POI by measuring the similarity of location representations, indicating a better performance. However, the algorithm could not adjust the size of the representations S_p (equals to the number of grids of the map), which is difficult to be applied to various POI recommendation models. Therefore, we conduct a lightweight network to learn new location representations with flexible sizes. Let e_i denotes the embedding of grid i , for each pair-wise grids i and j , we compute the similarity of the vectors using $e_i e_j$. The objective function is defined as follows:

$$L = \frac{1}{|P|} \frac{1}{|P|} \sum_{i=1}^{|P|} \sum_{j=1}^{|P|} (e_i e_j - S_i S_j)^2. \quad (2)$$

We use Adam to optimize the parameters. As Figure 1 (b) shows, the relationship of every S_i is mapped to e_i with a new dimension which is used to fit target POI recommendation models. Finally, we initialize the POI embedding by concatenating the location representations. Since the target POI recommendation models aggregate POIs by users, with the users visiting POIs in close regions, it is reasonable that the model can reach an earlier convergence if the POI embedding contains information about positional relationships.

Table 1: Performance of models on Toronto dataset

	NDCG@5	NDCG@10	NDCG@20	Hit@5	Hit@10	Hit@20
C-Context	0.0510	0.0759	0.1113	0.0857	0.1631	0.3044
C-Geohash	0.0699	0.0954	0.1297	0.1126	0.1919	0.3292
C-Geoinfluence	0.0547	0.0817	0.1147	0.0939	0.1787	0.3101
C-Proposed	0.1292	0.1658	0.2028	0.1963	0.3096	0.4573
S-Context	0.2014	0.2417	0.2839	0.2934	0.4186	0.6862
S-Geohash	0.2795	0.3232	0.3612	0.3883	0.5236	0.6742
S-Geoinfluence	0.2044	0.2495	0.2926	0.3022	0.4419	0.6127
S-Distinterval	0.2170	0.2595	0.3008	0.3155	0.4477	0.6114
S-Proposed	0.2858	0.3308	0.3700	0.4050	0.5452	0.7011

3 RESULTS

Dataset and Metrics. We use Yelp Challenge Round 13 dataset and select a subset of data in Toronto city due to its larger size. We adopt two common metrics: Hit@ k and NDCG@ k , to evaluate model recommendation performance. To measure the convergence rate of the models, we measure average regret in terms of Hit@10 [2]:

$$\frac{1}{T} \sum_{t=1}^T \text{Hit@10}(\bar{w}) - \text{Hit@10}(w^*), \quad (3)$$

where $\bar{w}$ is the base model, w^* is comparative models, and T is the number of epochs.

Baselines. We evaluated all methods on two widely used models: Collaborative filtering (C-) and Sequential model (S-) [3], we utilize every users' the most recent 50 check-ins as inputs of all S-models. (1) **Context**¹ [8] bridges neighborhood POIs in close area. (2) **Geohash**¹ [1] encodes POIs based on their locations on map. (3) **GeoInfluence**¹ [5] computes the relative locations using a standard normal distribution. (4) **DistInterval**² [4] is a sequential model that calculates the distance interval within a sequence.

Results. Table 1 shows that Proposed method outperforms all comparative methods on all metrics, indicating that the proposed location representation can improve the model performance. In particular, the improvement on C- models is more significant than on S- models, which may be caused by the larger amount of parameters of the C- models. Figure 2 presents the convergence efficiency of all S- models. The accuracy increases fast in early epochs (Proposed reaches 10% improvement compared to Geohash between epoch@50 to 150), while almost stable after 150 epochs. Taking the Proposed as the base model, the average regret of Context, Geohash, Geoinfluence, and Distinterval in the first 150 epochs are 0.14, 0.11, 0.04, and 0.10, respectively, with Proposed the fastest converge rate.

4 CONCLUSION

This work proposed a better location representation of POIs, which contains the positional relationship of POIs, to improve the model performance while accelerating the convergence rate. A lightweight network was designed to process the representation to be easily applied to various POI recommendation models.

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¹We only keep POI-location related part of these models.

²We replace the model's time interval with distance interval.

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